BMC Ophthalmology

RESEARCH

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Machine learning-based individualized prediction: risk assessment of retinopathy in preterm infants at high altitude

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Abstract

Background Retinopathy of prematurity (ROP) is a leading cause of childhood blindness. Chronic hypoxia at high altitude may create a unique risk profile and potentially influence the onset and progression of ROP. However, predictive tools specifically designed for preterm infants in high-altitude settings remain limited. This study aimed to develop and validate machine learning (ML)-based models for individualized ROP risk prediction in this population.

Methods We retrospectively collected clinical data from 2,138 preterm infants who underwent standardized ROP screening at Qinghai Red Cross Hospital between May 2014 and May 2025. The cohort was divided into a training set ($n=1,496$) and an independent testing set ($n=642$) in a 7:3 ratio. The prediction timepoint (t0) was defined as immediately before the first scheduled ROP fundus screening. Candidate predictors were initially screened using univariate analysis, followed by temporal plausibility assessment and LASSO logistic regression (λ_min), resulting in 24 predictors for subsequent modeling. Nine ML algorithms, logistic regression, decision tree, random forest, XGBoost, LightGBM, support vector machine (SVM), Gaussian Naive Bayes (GaussianNB), multilayer perceptron (MLP), and TabNet, as well as a soft-voting ensemble (Random Forest + XGBoost + LightGBM) were trained using five-fold cross-validation and tuned via Bayesian optimization. Model performance was then evaluated on the independent testing set.

Results In the independent testing set, the random forest model achieved the highest discrimination (AUC = 0.850, 95% CI: 0.795–0.906; PR-AUC = 0.607). XGBoost and the soft-voting ensemble demonstrated similar AUCs (both 0.845), with XGBoost providing the best overall classification balance (F1-score = 0.622; MCC = 0.573).

Conclusions ML models based on routinely available pre-screening clinical variables can effectively stratify ROP risk in preterm infants at high altitude. Anchoring predictions to the timepoint immediately before the first scheduled ROP examination enhances clinical interpretability and temporal validity, potentially enabling more efficient allocation of screening resources in this setting.

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Keywords Retinopathy of prematurity, Machine learning, Predictive model, LASSO regression, Random forest, High-altitude regions, SHAP analysis

Introduction

Retinopathy of prematurity (ROP) is characterized by retinal neovascularization in premature and low birth weight (BW) infants, with abnormal retinal vascular development as the central pathological feature. This condition can lead to tractional retinal detachment, making ROP one of the leading causes of childhood blindness [1, 2]. Under normal circumstances, retinal vessels fully develop between 36 and 40 weeks of gestation. In premature infants, however, early exposure to the extrauterine environment can disrupt retinal vascularization, creating ischemic and hypoxic regions. This disruption may trigger abnormal proliferation of retinal neovascularization, hemorrhage, and fibrous tissue formation due to excessive expression of vascular endothelial growth factor (VEGF), ultimately resulting in retinal detachment and permanent vision loss. According to a recent systematic review, the global prevalence of ROP among screened premature infants is approximately 32%, with severe ROP accounting for 7.5% of cases [3]. Longitudinal studies report that 50% of untreated threshold ROP cases progress to retinal detachment, 90% of which ultimately lead to blindness [3]. Furthermore, children with severe ROP exhibit a 20–30% incidence of cerebral palsy and a 40–50% prevalence of cognitive impairment, rates significantly higher than those observed in ROP-free premature infants [4].

Currently, ROP management relies primarily on regular fundus screening to identify threshold lesions that require timely laser photocoagulation or anti-VEGF therapy [5]. However, this approach faces two major challenges in practice. First, the rate of standardized fundus screening for high-risk premature infants remains relatively low in low- and middle-income countries, reaching only 20–30%, which is far below the 90% target recommended by the World Health Organization. Second, the commonly applied screening criteria, based on gestational age (GA) and BW, provide broad coverage but generate a substantial screening burden. This can lead to delayed diagnoses, missed cases, and increased pressure on limited clinical resources [6–8]. Furthermore, traditional ROP risk factors, such as low GA, low BW, and prolonged oxygen therapy, have limited predictive value for individualized risk stratification [9, 10]. These limitations underscore the urgent need for a more precise, data-driven risk evaluation strategy to efficiently identify high-risk infants, optimize allocation of medical resources, and ensure timely interventions.

The emergence and application of machine learning (ML) offer new opportunities to address the challenges described above. ML enables the analysis of complex, high-dimensional clinical data, allowing the identification of subtle patterns and interactions that are difficult to detect using traditional methods. This capability holds promise for the development of more robust ROP predictive models [11–13]. Despite preliminary studies demonstrating the feasibility of ML, several challenges and unresolved issues remain. Many existing models are constructed using limited clinical characteristics, often overlooking the potential impact of perinatal complications, such as bronchopulmonary dysplasia (BPD) and perinatal asphyxia (PA), as well as metabolic factors. Additionally, there is insufficient evidence supporting the generalizability of these models across different populations, particularly among those residing in unique environments, such as high-altitude regions [14, 15]. Chronic hypoxia in high-altitude areas, including Qinghai and Tibet, may shape distinct ROP risk profiles, raising uncertainty about the applicability of screening models developed for populations at lower altitudes. To address these gaps and improve clinical outcomes, it is therefore critical to develop customized ML predictive models specifically for populations chronically exposed to high-altitude hypoxia.

To address the gaps outlined above, this study aimed to develop and validate a ML-based predictive model for ROP using a retrospective cohort of premature infants residing at high altitude. Specifically, our study sought to: (1) identify independent risk factors for ROP and construct a concise, efficient feature set, and (2) systematically evaluate and compare the predictive performance of multiple ML algorithms to determine the optimal approach for this dataset. We hypothesized that a ML model integrating multidimensional clinical data with environmental exposure information could enable early and precise identification of infants at high risk for ROP. The findings are expected to provide a data-driven tool for optimizing screening strategies, facilitating timely interventions, and ultimately improving visual outcomes in this vulnerable population.

Materials and methods

Study objects

This retrospective cohort study enrolled preterm infants hospitalized in the Neonatology Department of Qinghai Red Cross Hospital between May 2014 and May 2025. All included infants underwent standardized ROP fundus screening using the RetCam 3 wide-field digital imaging system. ROP diagnosis was made according to the diagnostic criteria outlined in the 2014 Guidelines for

Screening ROP in China [16]. Inclusion criteria were: (1) GA < 37 weeks; (2) completion of the ROP screening protocol; and (3) availability of complete clinical records. Exclusion criteria were: (1) severe congenital deformities or chromosomal abnormalities; (2) other significant ocular pathologies interfering with ROP assessment; (3) missing key variables in more than 30% of cases (e.g., oxygen therapy duration, BW); and (4) death or voluntary discharge prior to initial fundus screening.

Clinical data collection

Data were retrospectively extracted from the electronic medical record system of Qinghai Red Cross Hospital. Variables included: (1) baseline clinical information, such as gender, GA, BW, and Apgar scores; (2) perinatal factors, including premature rupture of membranes (PROM) and antenatal steroid use; (3) treatment information, such as respiratory support, nutritional strategies, and blood transfusions; (4) related complications, encompassing respiratory disorders (e.g., BPD and respiratory distress syndrome [RDS]), circulatory disorders (e.g., patent ductus arteriosus [PDA] and persistent pulmonary hypertension of the newborn [PPHN]), neurological disorders (e.g., intraventricular hemorrhage [ICH]), infectious conditions (e.g., sepsis), and metabolic disturbances; and (5) laboratory parameters, including complete blood count, liver and kidney function tests, inflammatory markers (e.g., procalcitonin [PCT]), metabolites (e.g., lactate and 25-hydroxyvitamin D3 [25(OH)D_3]), and blood gas analysis results.

Data preprocessing and feature screening

Based on ROP status (the dependent variable, binary classification), the dataset was stratified and randomly split at a 7:3 ratio into a training set (n = 1,496) and a testing set (n = 642) [17]. Feature selection was conducted on the training set as follows:

(1) Univariate analysis: A total of 59 candidate indicators were initially screened. Continuous variables were tested for normality and presented as mean ± standard deviation if normally distributed, or as median (interquartile range) if non-normally distributed. Between-group differences were assessed using the independent samples t-test for normally distributed variables and the Mann-Whitney U test for non-normally distributed variables. Categorical variables were compared using the chi-square test or Fisher's exact test, depending on expected cell counts.

(2) Clinical and multicollinearity assessment: Candidate predictors were further refined by integrating clinical judgment with multicollinearity evaluation, retaining variables with a variance inflation factor (VIF) < 5.

(3) Automated feature selection: Least Absolute Shrinkage and Selection Operator (LASSO) regression was applied for automated feature selection.

The optimal penalty parameter (λ) was determined using 10-fold cross-validation, and λ_min (minimizing the cross-validated deviance) was selected for downstream modeling [18, 19].

Missing data imputation and class imbalance handling

Missing data imputation

Prior to modeling, missing data were assessed (Table S1). No candidate predictor exceeded the predefined missingness threshold of 30%; consequently, all 59 candidate variables had ≤ 30% missing data. Among these, 24 variables contained at least one missing value, with the highest missingness observed for 25-hydroxyvitamin D [25(OH)D] (27.4%), followed by PCT, (4.17%). Missingness for all other variables was generally low (≤ 1.58%).

Missing values were imputed using multiple imputation by chained equations (MICE) in R (mice package, version 3.14.0). Predictive mean matching (PMM) was applied, generating m = 5 imputed datasets under a fixed random seed with a maximum of 50 iterations. Since imputation was performed prior to the train-test split, the first completed dataset was used for subsequent splitting and model development [20]. Detailed missing values are shown in Table S1.

Class imbalance handling

In this study, the class imbalance of ROP outcomes in the training set (11.0% ROP-positive) was addressed using SMOTENC (Synthetic Minority Over-sampling Technique for Nominal and Continuous features) [21].

Over-sampling was applied after LASSO-based feature selection (λ_min; 24 predictors) and prior to model training, using only data available within the prespecified prediction window (t0, immediately before the first scheduled ROP fundus screening). Specifically, the training feature matrix X_{train} was constructed using the 24 selected predictors and paired with the binary outcome y_{Train} (ROP-positive vs. ROP-negative).

Because the predictor set included categorical variables, SMOTENC, implemented in Python's imbalanced-learn library (random_state = 42), was employed. Categorical features were identified as integer-typed columns in Xtrain and their indices were passed to SMOTENC. This method generates synthetic ROP-positive samples by interpolating continuous predictors while appropriately handling nominal predictors, thereby minimizing the risk of creating invalid category values [22].

Before resampling, the training set included 1,496 infants, with a 0.11 proportion of ROP-positive cases. After SMOTENC, the training set increased to 2,662

infants, achieving an approximately balanced class distribution (ROP-positive proportion 0.50). The independent testing set was not resampled and retained its original distribution ($n = 642$, including 76 [11.8%] ROP-positive infants) to reflect real-world prevalence and support unbiased model evaluation.

Model evaluation and statistical test

Both internal and external validations were employed to comprehensively assess model performance. Internal validation was conducted using five-fold cross-validation on the training set, in which four folds were used for model fitting and the remaining fold for validation in each iteration, ensuring that each fold served as the validation set once. The mean area under the receiver operating characteristic curve (ROC-AUC) across the five folds was calculated as a robust measure of discriminative performance [23, 24]. This approach is particularly suitable for evaluating models with class-imbalanced outcomes and was also used as the objective function for Bayesian hyperparameter optimization, thereby helping to reduce overfitting and improve generalizability.

Model performance was evaluated using a multidimensional metric system, including: (i) discrimination metrics, such as ROC-AUC; (ii) calibration metrics, including the Brier score, which quantifies the agreement between predicted probabilities and observed outcomes; and (iii) clinical utility metrics, assessed via decision curve analysis (DCA) to estimate the net benefit across decision thresholds and evaluate potential clinical value in practice [25–27]. Descriptive statistics were used to summarize quantitative and categorical variables as appropriate. Differences in AUC performance across models on the independent testing set were compared using DeLong's test. Details of the Bayesian optimization settings and the optimal hyperparameter combinations for each algorithm are provided in Table S3.

ML

In this study, nine representative ML models encompassing diverse modeling approaches were constructed and compared to systematically identify the optimal algorithm for ROP prediction. The models included gradient boosting frameworks (XGBoost and LightGBM), kernel-based methods (support vector machine [SVM]), linear models (logistic regression), tree-based models (decision tree and random forest), probabilistic models (Gaussian Naive Bayes [GaussianNB]), and deep learning models (multilayer perceptron [MLP] and TabNet). To ensure objective and fair performance comparisons, all models were independently trained and tuned on the same training set.

Bayesian hyperparameter optimization was applied to reduce overfitting and improve model performance. For scikit-learn-implemented models, optimization was performed using BayesSearchCV from the scikit-optimize library (v0.9.0), with the mean five-fold cross-validation AUC as the objective function [28–30]. For TabNet, Bayesian optimization was conducted using gp_minimize. The optimal hyperparameter configurations for all models are summarized in Table S3.

To evaluate generalization performance, a completely independent testing set was used for external validation of the final models. Model performance was assessed using a comprehensive set of metrics, including ROC-AUC (with 95% CI), PR-AUC, accuracy, sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), F1-score, Brier score, and Matthews correlation coefficient (MCC). This systematic modeling framework, from data preprocessing and feature engineering to model training and rigorous validation, ensured reliable performance comparison across algorithms and reproducibility of the study workflow and results [31].

Data processing and statistical analysis

All statistical analyses and predictive model development were performed within a dedicated computational environment. R software (version 4.2.1) was used for variable selection, univariate analysis, and LASSO regression. ML model training, hyperparameter optimization, and evaluation were conducted using Python libraries, including scikit-learn, XGBoost, and LightGBM, within the Anaconda Navigator platform.

Results

Comparison of baseline characteristics

To evaluate the comparability of the datasets after stratified sampling, baseline characteristics were systematically compared between the training set ($n = 1{,}496$) and the testing set ($n = 642$), as presented in Table 1. No statistically significant differences were observed in the baseline variables listed in Table 1 (all $P > 0.05$), supporting the validity of the sampling strategy and the representativeness of the independent testing set. Importantly, the outcome distribution was similar between the two sets (ROP: 11.8% in the testing set vs. 11.0% in the training set; $P = 0.640$). Key predictors used in subsequent modeling also showed consistent distributions across the two datasets, including PSA (24.9% vs. 23.2%, $P = 0.421$), RF (46.3% vs. 49.9%, $P = 0.139$), and PA (18.1% vs. 18.4%, $P = 0.912$). Continuous variables likewise demonstrated comparable distributions, including FiO_2 (median 25.00 vs. 25.00, $P = 0.973$), GA (median 34.14 vs. 34.14 weeks, $P = 0.647$), and BW (median 1,960 vs. 1,950 g, $P = 0.824$). Collectively,

Table 1 Comparison of baseline characteristics between the training and independent testing sets. (key variables)

Characteristic	Training set ($n=1496$)	Testing set ($n=642$)	*p* value
Male, n (%)	844(56.4)	352(54.8)	0.528
Altitude category, n (%)			0.396
Low (< 1500 m)	3(0.2)	0(0)	
Middle (1500–2500 m)	824(55.1)	343(53.4)	
High (≥ 2500 m)	669(44.7)	299(46.6)	
ROP, n (%)	165(11)	76(11.8)	0.640
Predictors included in the final model			
Mode of delivery = Cesarean section, n (%)	1025(68.5)	430(67)	0.517
Premature rupture of membranes, n (%)	522(34.9)	226(35.2)	0.930
Antenatal corticosteroid exposure, n (%)	873(58.4)	385(60)	0.518
Respiratory failure, n (%)	746(49.9)	297(46.3)	0.139
Respiratory distress syndrome, n (%)	1079(72.1)	457(71.2)	0.696
Perinatal asphyxia, n (%)	275(18.4)	116(18.1)	0.912
Blood component transfusion, n (%)	586(39.2)	246(38.3)	0.747
Pulmonary surfactant administration, n (%)	347(23.2)	160(24.9)	0.421
Fraction of inspired oxygen (FiO_2), %, median (IQR)	25.00(23.00, 30.00)	25.00(23.00, 30.00)	0.973
Gestational age, weeks, median (IQR)	34.14(32.43, 35.43)	34.14(32.43, 35.57)	0.647
Birth weight, g, median (IQR)	1950.00(1600.00, 2250.00)	1960.00(1600.00, 2300.00)	0.824
Apgar score at 1 min, median (IQR)	8.00(7.00, 9.00)	8.00(6.00, 9.00)	0.496
Apgar score at 5 min, median (IQR)	9.00(8.00, 10.00)	9.00(8.00, 10.00)	0.571
Lactic acid, median (IQR)	2.60(1.80, 3.90)	2.60(1.80, 3.80)	0.733
Partial pressure of carbon dioxide, median (IQR)	36.80(31.70, 42.20)	36.15(31.00, 41.70)	0.179
Lymphocyte count, median (IQR)	3.50(2.70, 4.58)	3.40(2.60, 4.56)	0.568
Red blood cell count, median (IQR)	4.73(4.34, 5.08)	4.77(4.38, 5.14)	0.329
Hematocrit, median (IQR)	53.40(49.10, 57.70)	53.90(48.60, 57.70)	0.943
RDW-SD, median (IQR)	64.70(60.90, 69.70)	64.65(60.80, 70.10)	0.951
Platelet count, median (IQR)	247.00(202.00, 289.00)	243.00(199.00, 285.00)	0.214
Total bile acids, μmol/L, median (IQR)	9.20(6.10, 13.85)	9.00(5.90, 13.90)	0.778
Albumin, mean ± SD	30.55 ± 3.99	30.40 ± 3.99	0.412
Procalcitonin, median (IQR)	0.80(0.23, 3.68)	0.94(0.24, 4.24)	0.417
Serum 25-hydroxyvitamin D3, median (IQR)	23.28(18.50, 29.72)	23.60(18.90, 29.96)	0.463

Footnote: Data are presented as n (%) for categorical variables and as median (IQR) for continuous variables (as appropriate). p values were calculated using the chi-square test or Fisher's exact test for categorical variables and the independent-samples t-test or Mann–Whitney U test for continuous variables, depending on distribution

Abbreviations: SD, standard deviation; IQR, interquartile range; ROP, retinopathy of prematurity. (All other abbreviations are defined in Table S6)

these findings indicate strong inter-set balance in both baseline characteristics and key modeling variables, providing a reliable foundation for subsequent model training and evaluation of generalization performance.

To systematically identify potential predictors of ROP, clinical characteristics were compared between the ROP lesion group ($n=165$) and the non-lesion group ($n=1,331$) within the training set, as presented in Table S2A and Table S2B. Univariate analyses revealed significant differences between the two groups across multiple domains ($P<0.05$). Infants in the lesion group had markedly lower GA and BW, as well as lower Apgar scores at 1 and 5 min (all $P<0.001$), indicating that greater prematurity and poorer early postnatal condition were strongly associated with the occurrence of ROP.

Consistent with the oxygen-related pathophysiology of ROP, the lesion group demonstrated substantially higher respiratory support intensity and oxygen exposure, including a longer duration of oxygen therapy (O2D) and a higher fraction of inspired oxygen (FiO_2) (both $P<0.001$). In addition, the lesion group showed significantly higher rates of surfactant administration (PSA), invasive mechanical ventilation (IMV), and more severe respiratory-support profiles (all $P<0.001$). Interestingly, RF (1 = yes) was less frequent in the lesion group (29.7% vs. 52.4%, $P<0.001$; Table S2A and Table S2B). This inverse association observed in univariate analysis was interpreted cautiously, as it may reflect differences in clinical care pathways or correlations with other respiratory and oxygen-related indicators rather than a true protective effect. Several neonatal complications and perinatal factors also differed significantly between the two groups, including mode of delivery (MOD, $P<0.001$), PROM ($P=0.003$), antenatal corticosteroid exposure

(ACS, $P = 0.027$), PA ($P < 0.001$), RDS ($P < 0.001$), BPD ($P < 0.001$), necrotizing enterocolitis (NEC, $P < 0.001$), hyperbilirubinemia (HB, $P < 0.001$), intracranial hemorrhage (ICH, $P = 0.007$), and other clinically relevant morbidities such as persistent PPHN ($P = 0.037$).

Laboratory indicators also demonstrated characteristic differences between the two groups. The lesion group exhibited higher blood lactate (cLac, $P < 0.001$), elevated total bile acids (TBA, $P < 0.001$), increased PCT ($P = 0.012$), and higher 25(OH)D levels ($P < 0.001$). In contrast, nutritional and hematologic markers, including albumin (ALB, $P < 0.001$), hemoglobin (HGB, $P < 0.001$), red blood cell count (RBC, $P < 0.001$), and hematocrit (HCT, $P = 0.004$), were significantly lower in the lesion group. Platelet count (PLT, $P = 0.005$) and RDW-SD ($P = 0.049$) also differed significantly between groups.

In contrast, several variables showed no statistically significant differences between the two groups, including gender ($P = 0.286$), multiple maternal comorbidities (e.g., gestational diabetes mellitus [GDM], intrahepatic cholestasis of pregnancy [ICP], and gestational hypertension [GH]), sepsis ($P = 0.124$), and certain physiological indicators such as arterial oxygen saturation (SaO_2, $P = 0.760$) and corrected GA (CGA, $P = 0.853$). Collectively, the univariate results presented in Table S2A and Table S2B identified multidimensional candidate predictors spanning prematurity severity, oxygen and respiratory support intensity, perinatal exposures and complications, as well as metabolic, inflammatory, and hematologic biomarkers. These findings provided a strong rationale for subsequent feature selection and the development of machine-learning prediction models.

Table 2 Predictors selected by LASSO regression for ROP risk modeling (training set; λ_min)

Predictor	Type	Unit / Coding	Coefficient
Mode of delivery	Binary	Vaginal delivery = 0; Cesarean section = 1	0.450796
Premature rupture of membranes	Binary	Yes = 1; No = 0	0.552631
Antenatal corticosteroid exposure	Binary	Yes = 1; No = 0	0.012085
Respiratory failure	Binary	Yes = 1; No = 0	-1.268028
Respiratory distress syndrome	Binary	Yes = 1; No = 0	-0.718217
Perinatal asphyxia	Binary	Yes = 1; No = 0	1.482549
Blood component transfusion	Binary	Yes = 1; No = 0	-0.059801
Pulmonary surfactant administration	Binary	Yes = 1; No = 0	1.579772
Fraction of inspired oxygen (FiO_2)	Continuous	%	0.242981
Gestational age	Continuous	weeks	-0.438951
Birth weight	Continuous	g	-0.571620
Apgar score at 1 min	Continuous	1-minute	0.076668
Apgar score at 5 min	Continuous	5-minute	0.057330
Lactic acid	Continuous	mmol/L	0.015761
Partial pressure of carbon dioxide	Continuous	%	0.123835
Lymphocyte count	Continuous	$*10^9$/L	-0.075769
Red blood cell count	Continuous	$*10^{12}$/L	-0.039220
Hematocrit	Continuous	%	-0.121615
RDW-SD	Continuous	fL	-0.104057
Platelet count	Continuous	$*10^9$/L	-0.128437
Total bile acids	Continuous	μmol/L	0.130276
Albumin	Continuous	g/L	0.322398
Procalcitonin	Continuous	ng/mL	-0.125702
Serum 25-hydroxyvitamin D3	Continuous	ng/mL	0.200755

Footnote: Predictors were selected using LASSO regression at λ_min. LASSO coefficients are shown for feature selection and do not directly represent odds ratios

Feature screening based on LASSO regression

Based on the training set ($n = 1{,}496$), candidate predictors were initially screened using univariate analyses ($P < 0.05$), yielding 40 variables. Variables with potential temporal-order ambiguity relative to the intended prediction timepoint (t_0, defined as immediately prior to the first scheduled ROP examination) were subsequently excluded following clinical review, leaving 27 variables for penalized regression. LASSO logistic regression with an L1 penalty was then applied for automated feature selection and dimensionality reduction. Continuous predictors were standardized before modeling. The penalty parameter (λ) was determined using 10-fold cross-validation, and λ_min (which minimizes the cross-validated deviance) was selected for downstream machine-learning modeling. Consequently, 24 predictors with non-zero penalized coefficients were retained (Table 2).

The selected feature set encompassed multiple domains, including maturity indicators (GA and BW), perinatal and delivery factors (e.g., MOD, PROM, ACS, and Apgar scores), early respiratory and oxygen-related indicators prior to t_0 (e.g., surfactant administration [PSA], RDS, RF, FiO_2, and PCO_2), and laboratory biomarkers reflecting metabolic, inflammatory, and hematologic status (e.g., blood lactate [cLac], TBA, ALB, PCT, 25(OH)D, and blood-count indices). Notably, RF was retained with a negative penalized coefficient under λ_min. This directionality should be interpreted cautiously, as it likely reflects model-conditional contributions influenced by correlations and interactions with other oxygen- and respiratory-related variables within the same pre-screening window (e.g., PSA and FiO_2), rather than evidence of a causal protective effect. Overall, LASSO reduced the feature space from 59 initial candidate

variables to a concise set of 24 predictors, facilitating parsimonious model construction while mitigating the risk of overfitting. It should be noted that LASSO coefficients are reported solely to document the feature selection process and should not be interpreted as unbiased effect estimates (e.g., odds ratios) or as evidence of causal relationships.

Model performance evaluation

In this study, model performance was evaluated using internal validation on the training set (five-fold cross-validation) and external validation on an independent testing set. The cross-validated AUC values obtained from the training set are summarized in Table 3A and Table 3B, while comprehensive performance metrics for the testing set are presented in Table 3A and Table 3B, including ROC-AUC (with 95% CI), PR-AUC, accuracy, sensitivity, specificity, PPV, NPV, F1-score, and MCC.

Overall, the models demonstrated moderate-to-good discriminative performance on the testing set, with AUC values ranging from 0.764 to 0.850 (Table 3A and Table 3B). The random forest model achieved the highest ROC-AUC (0.850, 95% CI: 0.795–0.906) and a PR-AUC of 0.607, along with an accuracy of 0.897, sensitivity of 0.618, specificity of 0.935, PPV of 0.560, NPV of 0.948, F1-score of 0.588, and MCC of 0.530. The XGBoost model and the soft-voting ensemble demonstrated comparable discriminative ability (both AUC = 0.845). Notably, XGBoost achieved the best overall balance across classification metrics, with the highest F1-score (0.622) and MCC (0.573) .

Notably, several high-capacity models achieved very high AUC values during five-fold cross-validation on the training set, but their performance declined on the independent testing set. This discrepancy suggests potential overfitting and highlights the importance of independent validation when evaluating predictive models in a single-center cohort.

A soft-voting ensemble model was constructed by integrating the three best-performing base learners (Random Forest, XGBoost, and LightGBM). Each base learner outputs the predicted probability of the positive class, and the final ensemble probability is calculated as a weighted average of these probabilities:

$$Pfinal(class = i) = \sum (wk * Pk(class = i))$$

where Pfinal (class = i) represents the final probability that a sample belongs to Class I as predicted by the ensemble model; wk denotes the weights assigned to the kth base

Table 3A Discrimination and classification performance of the machine-learning models in the training set (5-fold cross-validation) and the independent testing set (Part A)

Model	Training AUC (5-fold CV)	AUC (testing, 95% CI)	PR-AUC	Accuracy	Sensitivity	Specificity
RandomForest	0.980	0.850 (0.795–0.906)	0.607	0.897	0.618	0.935
Soft-voting ensemble	0.986	0.845 (0.789–0.902)	0.605	0.910	0.592	0.952
XGBoost	0.987	0.845 (0.789–0.901)	0.609	0.913	0.605	0.954
LightGBM	0.989	0.840 (0.783–0.897)	0.613	0.914	0.553	0.963
Multilayer perceptron	0.899	0.833 (0.775–0.891)	0.445	0.816	0.684	0.834
TabNet	1.000	0.805 (0.744–0.866)	0.498	0.875	0.618	0.910
DecisionTree	0.939	0.793 (0.731–0.856)	0.492	0.868	0.579	0.906
Support vector machine	0.984	0.789 (0.727–0.852)	0.396	0.880	0.303	0.958
Gaussian naive Bayes	0.866	0.788 (0.726–0.851)	0.456	0.765	0.579	0.790
Logistic regression	0.848	0.764 (0.699–0.829)	0.375	0.746	0.618	0.763

Footnote: "Training AUC" refers to the mean area under the receiver operating characteristic curve (ROC-AUC) obtained by 5-fold cross-validation in the training set. Testing performance was evaluated on a completely independent testing set and is reported as ROC-AUC with 95% confidence interval (CI), precision–recall AUC (PR-AUC), accuracy, sensitivity, and specificity. Tables 3A and 3B present complementary performance metrics for the same set of models and were separated only for readability due to table length

Abbreviations: AUC, area under the curve; ROC-AUC, area under the receiver operating characteristic curve; PR-AUC, area under the precision–recall curve; CI, confidence interval

Table 3B Predictive values and overall classification balance of the machine-learning models in the training set (5-fold cross-validation) and the independent testing set (Part B)

Model	Training AUC (5-fold CV)	AUC (testing, 95% CI)	PPV	NPV	F1-score	MCC
RandomForest	0.980	0.850 (0.795–0.906)	0.560	0.948	0.588	0.530
Soft-voting ensemble	0.986	0.845 (0.789–0.902)	0.625	0.946	0.608	0.557
XGBoost	0.987	0.845 (0.789–0.901)	0.639	0.947	0.622	0.573
LightGBM	0.989	0.840 (0.783–0.897)	0.667	0.941	0.604	0.560
Multilayer perceptron	0.899	0.833 (0.775–0.891)	0.356	0.952	0.468	0.399
TabNet	1.000	0.805 (0.744–0.866)	0.480	0.947	0.540	0.475
DecisionTree	0.939	0.793 (0.731–0.856)	0.454	0.941	0.509	0.438
Support vector machine	0.984	0.789 (0.727–0.852)	0.489	0.911	0.374	0.323
Gaussian naive Bayes	0.866	0.788 (0.726–0.851)	0.270	0.933	0.368	0.274
Logistic regression	0.848	0.764 (0.699–0.829)	0.260	0.937	0.366	0.274

Footnote: "Training AUC" refers to the mean ROC-AUC obtained by 5-fold cross-validation in the training set. Testing performance was evaluated on a completely independent testing set and is reported as ROC-AUC with 95% CI, positive predictive value (PPV), negative predictive value (NPV), F1-score, and Matthews correlation coefficient (MCC). Tables 3A and 3B report different metrics for the same models and were split into two parts for readability

Abbreviations: AUC, area under the curve; CI, confidence interval; PPV, positive predictive value; NPV, negative predictive value; MCC, Matthews correlation coefficient

model, derived from standardised coefficients based on the AUC values (five-fold cross-validation AUC on the training set) of each model; and Pk (class = i) represents the probability that a sample belongs to Class I predicted by the kth base model.

To prevent information leakage from the testing set during ensemble construction, model weights were determined by normalizing the five-fold cross-validated AUC values obtained from the training set for the selected base learners. On the independent testing set, the soft-voting ensemble achieved an AUC of 0.845 (95% CI: 0.789–0.902) and a PR-AUC of 0.605, demonstrating a favorable overall balance across evaluation metrics (Table 3A and Table 3B).

To determine whether differences in discriminative performance were statistically significant, pairwise comparisons of ROC-AUC values on the testing set were conducted using DeLong's test (Table S4). The random forest model significantly outperformed logistic regression ($\Delta AUC = 0.086$, $P = 0.001$) and the decision tree ($\Delta AUC = 0.057$, $P = 0.014$), and also demonstrated a significantly higher AUC than TabNet ($\Delta AUC = 0.045$, $P = 0.035$). In contrast, no significant differences in AUC were observed between random forest and XGBoost ($\Delta AUC = 0.005$, $P = 0.606$) or between random forest and the voting ensemble ($\Delta AUC = 0.005$, $P = 0.573$).

Model performance was further evaluated using ROC curves, calibration plots, and DCA (Figs. 1, 2 and 3) to assess discrimination, probability calibration, and potential clinical utility. The ROC curves (Fig. 1) visually confirmed the relative discriminative performance across models, consistent with the AUC estimates reported in Table 3A and Table 3B.

For calibration (Fig. 2), the calibration curves were generally located below the ideal diagonal line, suggesting systematic overestimation of predicted risk across probability bins on the independent testing set (i.e., predicted probabilities tended to be higher than the observed event rates). This pattern may partly reflect differences in class prior distributions between the resampled training data and the naturally imbalanced testing data. Calibration performance should therefore be interpreted cautiously because (i) some models underwent additional post-calibration, and (ii) the training data were resampled, whereas the testing set retained the natural prevalence. To quantitatively assess calibration error, we also calculated the Brier score on the testing set, where lower values indicate better overall calibration and accuracy of probabilistic predictions. The Brier scores were 0.078 for random forest, 0.074 for XGBoost, 0.071 for LightGBM, and 0.074 for the voting ensemble, indicating broadly comparable calibration performance among these top models, with a slight advantage for LightGBM (Table S5).

Decision-curve analysis (Fig. 3) indicated that the better-performing models provided higher net benefit than

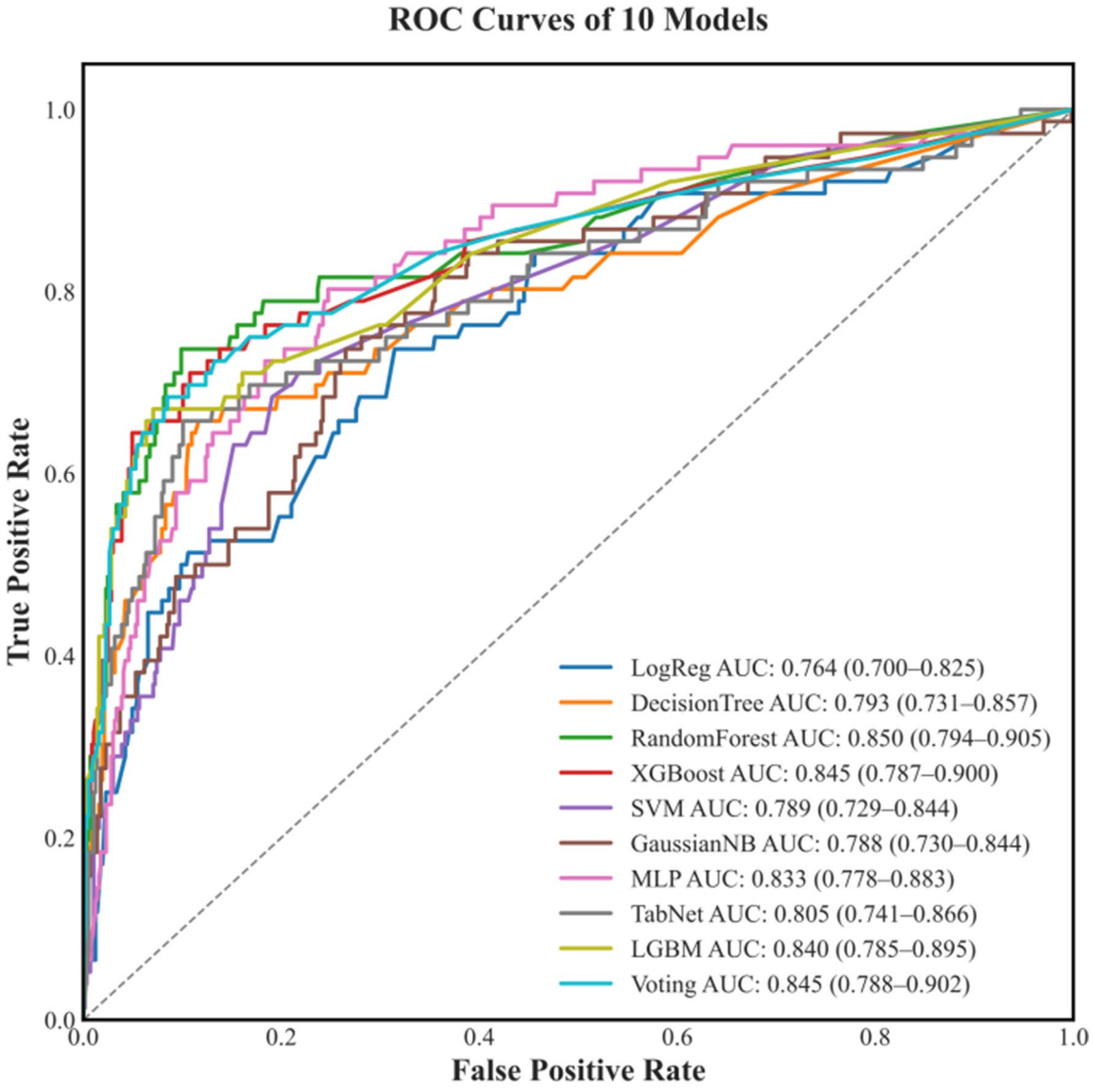


Fig. 1 Receiver operating characteristic (ROC) curves of the 10 predictive models on the independent testing set. ROC curves are shown for the 10 models; ROC-AUCs with 95% confidence intervals are provided in the legend. The grey dashed diagonal indicates no-discrimination (AUC = 0.5). The soft-voting ensemble combined Random Forest, XGBoost, and LightGBM using a weighted average of predicted probabilities, with weights normalized from the mean 5-fold cross-validated AUCs obtained in the training set

both the "treat-all" and "treat-none" strategies across clinically relevant threshold probabilities, supporting their potential utility for risk stratification prior to the first scheduled ROP examination.

Model recommendation. Considering (i) the highest ROC-AUC on the independent testing set, (ii) competitive PR-AUC and a clinically meaningful balance of sensitivity and specificity, (iii) no significant AUC disadvantage relative to XGBoost or the voting ensemble according to DeLong's test, and (iv) the interpretability and robustness of tree-based ensembles, the random forest model was selected as the recommended model for subsequent SHAP-based interpretation and potential clinical translation (Table 3A and Table 3B; Table S4). Although XGBoost and the voting ensemble achieved comparable AUC values and slightly higher F1-score and MCC, and LightGBM showed a marginally lower Brier score, the random forest maintained competitive calibration (Brier score = 0.078) while providing the highest ROC-AUC and a clinically acceptable balance of errors.

Model interpretability analysis

To enhance clinical interpretability, we applied the Shapley Additive Explanations (SHAP) framework to the random forest model, which achieved the highest AUC on the independent testing set (Table 3A and Table 3B) and was selected as the recommended model. SHAP values

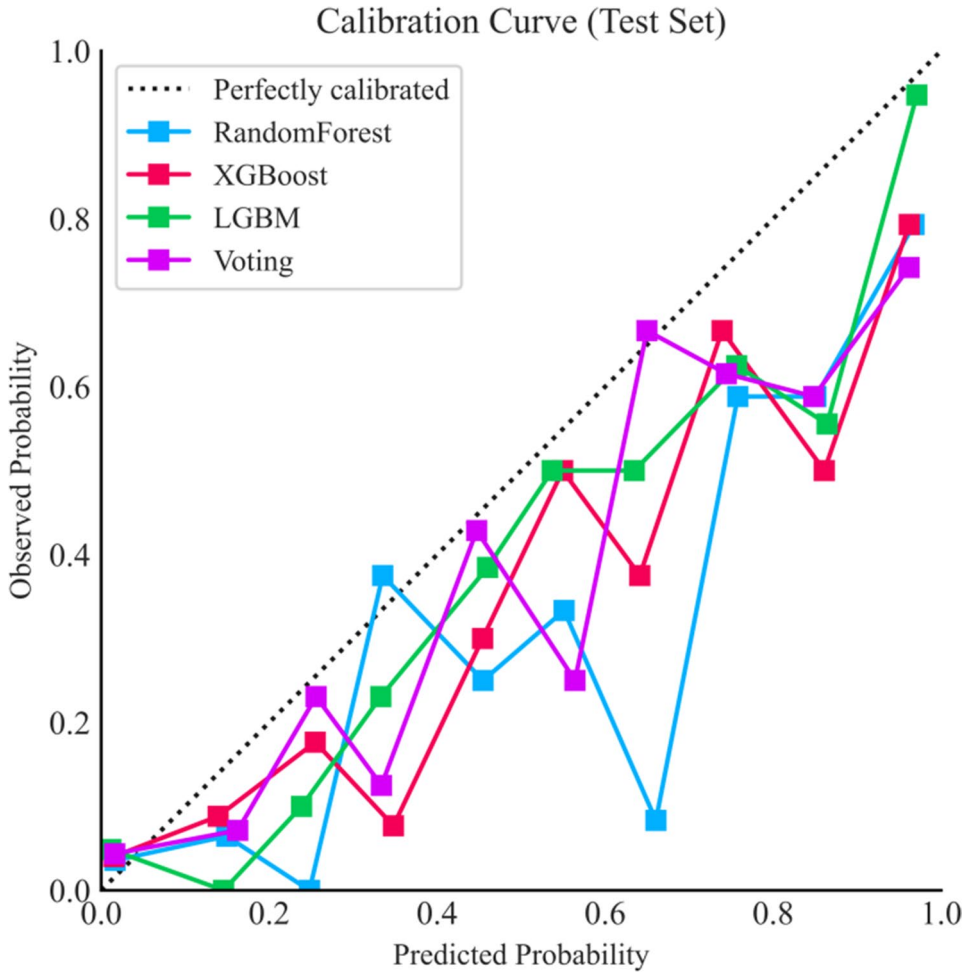


Fig. 2 Calibration curves of the top-performing models on the independent testing set. Calibration curves compare predicted probabilities with observed event proportions. The dotted diagonal line indicates perfect calibration. Random Forest, XGBoost, LightGBM, and the soft-voting ensemble are shown. Deviations from the diagonal reflect miscalibration (over- or under-estimation of risk) across probability ranges

were calculated using the TreeSHAP algorithm, specifically designed for tree-based ensemble models. To balance computational efficiency with explanation stability, SHAP estimation was performed on a randomly selected subset of 100 testing samples with a fixed random seed. Global feature importance was quantified using the mean absolute SHAP values, and feature effect directions were visualized via SHAP summary plots, enabling assessment of how individual predictor values contribute to increases or decreases in ROP risk predictions.

Figure 4a presents the global feature importance of the recommended random forest model, measured by mean absolute SHAP values. All explanations were restricted to predictors available prior to t_0 (defined as immediately before the first scheduled ROP fundus screening). The six most influential predictors, in descending order, were pulmonary surfactant administration (PSA), respiratory failure (RF), GA, PA, BW, and FiO_2. This pattern indicates that, within the pre-screening prediction framework, model risk stratification is primarily driven by early respiratory disease severity and oxygen-related exposures, together with neonatal maturity indicators, findings that are clinically plausible for ROP risk assessment.

The SHAP summary plot (Fig. 4b) further illustrates the directionality of feature effects. Higher values or presence of PSA, PA, and FiO_2 were predominantly associated with positive SHAP values, indicating an increased predicted probability of ROP. In contrast, higher GA and BW were mainly associated with negative SHAP values, reflecting protective contributions consistent with established epidemiological evidence. Notably, higher RF values (RF = 1) were mainly associated with negative SHAP values. This should be interpreted as a model-conditional contribution rather than a causal protective effect. In tree-based ensembles, SHAP quantifies the marginal contribution of a feature conditional on other correlated predictors, such as PSA and FiO_2. Therefore, the observed negative SHAP

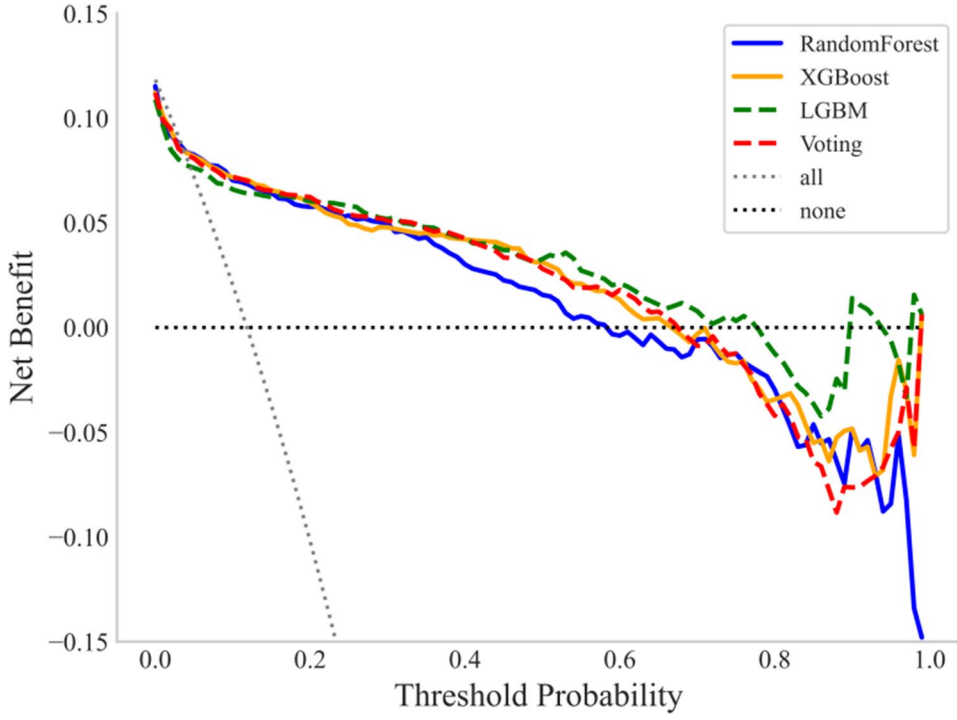


Fig. 3 Decision-curve analysis (DCA) of the top-performing models on the independent testing set. Decision curves show the net benefit across a range of threshold probabilities for Random Forest, XGBoost, LightGBM, and the soft-voting ensemble. The "treat-all" and "treat-none" strategies are shown as reference lines. For a given threshold probability, a model provides clinical utility when its net benefit is higher than both reference strategies

direction for RF may reflect feature redundancy, interactions, or care-pathway-dependent documentation within the pre-screening window, and warrants further validation in external cohorts.

Finally, the dominant predictors identified by SHAP (PSA, RF, FiO_2, GA, BW, and PA) largely overlapped with the variables retained during LASSO-based feature selection (λ_min), supporting the robustness and clinical plausibility of the feature screening strategy and enhancing the transparency of the final model.

Discussion

In this study, we developed and validated ML-based models to predict ROP risk in preterm infants at high altitude using routinely collected clinical data. To enhance clinical usability and ensure temporal validity, model predictions were explicitly anchored to a prespecified and actionable timepoint (t_0, immediately prior to the first scheduled ROP fundus screening), and all predictors were restricted to information available before t_0. Using a stepwise feature-selection strategy, including univariate screening, temporal plausibility assessment, and LASSO logistic regression with λ_min, we derived a concise set of 24 predictors for downstream modeling. In independent testing, overall model discrimination was good, with AUC values ranging from 0.764 to 0.850. The random forest model achieved the highest AUC (0.850), while XGBoost and the soft-voting ensemble demonstrated comparable discrimination (AUC = 0.845) along with a more favorable overall classification balance (e.g., F1-score and MCC for XGBoost), supporting their potential utility for pre-screening risk stratification and optimization of screening allocation.

Our findings further refine prior evidence on the multidimensional determinants of ROP. Beyond traditional maturity-related factors, such as GA and BW, the selected feature set underscores the importance of perinatal exposures (e.g., antenatal corticosteroid use and PROM), early respiratory status and oxygen-related indicators prior to the first examination (e.g., RF, RDS, surfactant administration, and FiO_2), and laboratory markers reflecting inflammation, nutrition, and metabolic status (e.g., PCT, ALB, TBA, and 25(OH)D). Collectively, these results suggest that ROP risk in high-altitude settings is shaped by the complex interplay among immaturity, early respiratory and oxygen dynamics, and systemic metabolic-inflammatory conditions, providing a rationale for more targeted monitoring strategies before the first scheduled ROP examination [32–36].

In recent years, numerous predictive models have been developed using routinely collected clinical data, such as BW, GA, postnatal weight gain, duration of oxygen therapy, and number of blood transfusions. Well-known models include ROPScore, PW-ROP, WINROP, G-ROP, and DIGI-ROP [37]. With advances in artificial intelligence, ML (e.g., XGBoost) and deep learning-based

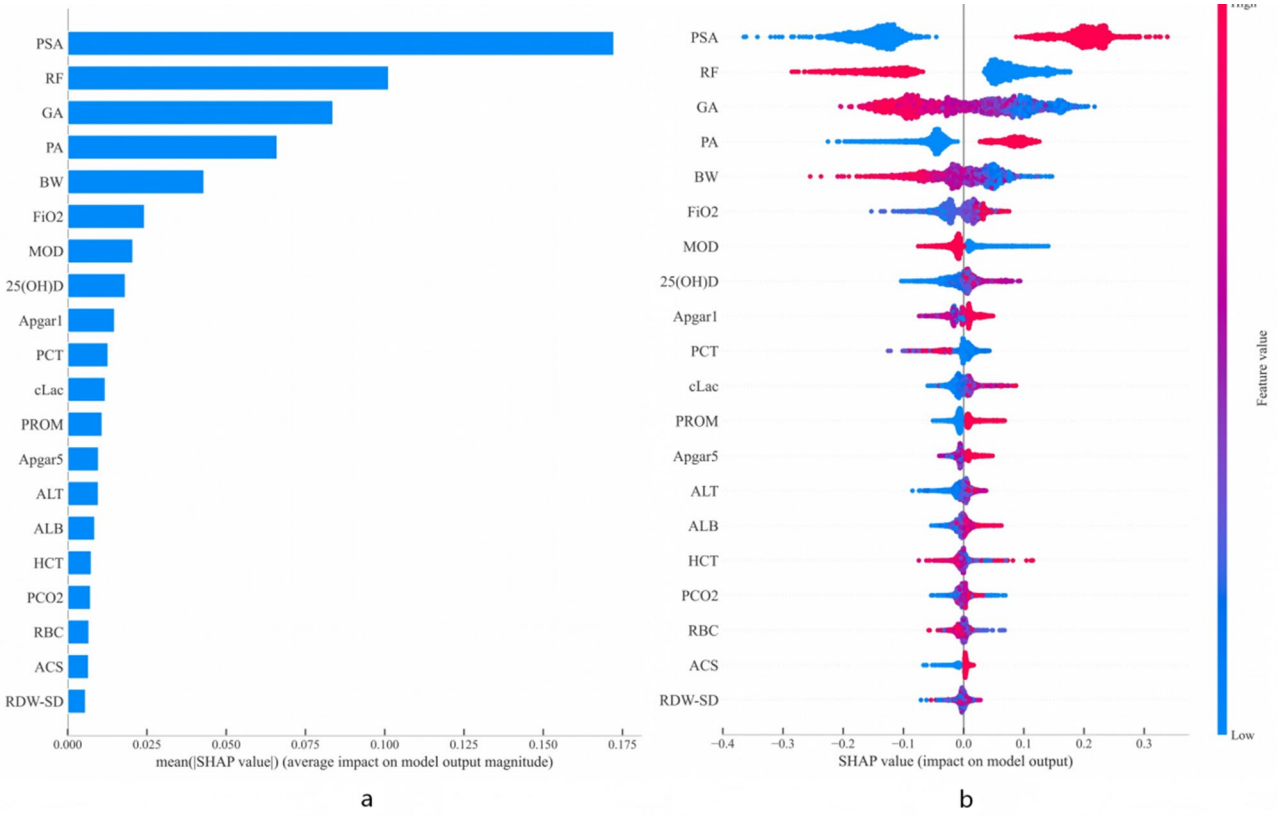


Fig. 4 SHAP-based interpretation of the recommended Random Forest model. (**a**) Global feature importance ranked by the mean absolute SHAP value, (average impact on model output magnitude). (**b**) SHAP summary plot showing the distribution of SHAP values for each feature; each point represents an individual infant, colored by the feature value, (low to high). Positive SHAP values indicate increased predicted risk of ROP, whereas negative values indicate decreased predicted risk. All features were restricted to information available prior to the prespecified prediction timepoint, (t_0, immediately before the first scheduled ROP screening)

models have demonstrated superior predictive performance compared with traditional approaches. Notably, this study is the first to demonstrate that ML predictive models can achieve superior discrimination for ROP risk in preterm infants at high altitude relative to conventional screening criteria. Moreover, the application of the SHAP interpretability framework enables effective alignment between model predictions and clinical knowledge, while also revealing the nonlinear effects of key predictors [38, 39]. Together, these innovations provide a methodological foundation for translating complex ML models into reliable, clinician-trusted decision-support tools.

Our findings indicate that the proposed ML framework could support clinical decision-making for ROP screening in preterm infants at high altitude, particularly for risk stratification and optimization of screening-resource allocation. To ensure clinical interpretability and temporal validity, predictions were anchored to a prespecified and actionable timepoint (t_0, immediately prior to the first scheduled ROP fundus screening), using only information available before t_0. Within this intended use scenario, integrating maturity indices (e.g., GA and BW) with early respiratory and oxygen-related indicators (e.g., PSA, FiO_2, and PA) and routinely available laboratory features may help identify infants who would benefit from closer ophthalmic surveillance and more frequent follow-up. Importantly, the model is not intended to replace guideline-based screening criteria; rather, it is designed as a complementary, data-driven tool to support risk-based prioritization before the first examination in similar care settings.

Several limitations should be acknowledged. First, this was a retrospective, single-center study; thus, the observed model performance may partly reflect local population characteristics and care pathways, and high internal discrimination should not be interpreted as evidence of universal transportability. Second, although an independent internal testing set was used, external validation in geographically distinct cohorts is required to evaluate generalizability across institutions and altitudes. Third, probability calibration on the testing set showed systematic deviation, and Brier scores indicated only moderate probabilistic accuracy; consequently, predicted probabilities may require recalibration or local threshold adjustment before clinical implementation. Fourth, missing data were imputed prior to the train-test split (with

no variable exceeding 30% missingness), which could introduce optimistic bias due to information sharing between datasets; test-set performance should therefore be interpreted cautiously and confirmed using leakage-free imputation workflows. In addition, oversampling (SMOTENC) was applied prior to cross-validation within the training set, which may have introduced optimistic bias in internal validation; future studies should perform resampling within each training fold and validate models in external cohorts. Finally, because the analysis relied solely on routinely collected clinical variables, mechanistic conclusions cannot be drawn, and biological experiments were not performed to validate pathways potentially implicated by certain biomarkers (e.g., TBA).

Future work could proceed along three main directions. First, multicenter prospective validation with standardized definitions and harmonized data collection is needed to assess generalizability and clinical impact across diverse settings. Second, incorporating dynamic physiological trajectories, such as longitudinal oxygen exposure patterns and cardiorespiratory monitoring signals, may enhance predictive timeliness and robustness beyond static baseline features. Third, because some predictors (e.g., RF) reflect clinician-assigned diagnoses and may be influenced by documentation heterogeneity and local care pathways, the directionality of model-based explanations should be interpreted with caution. External validation and mechanism-focused studies integrating molecular or experimental techniques are warranted to clarify the biological relevance of selected biomarkers (e.g., TBA) and to strengthen interpretability.

Conclusion

In conclusion, this study developed and internally validated ML-based models to predict the risk of ROP in preterm infants at high altitude using routinely collected clinical variables. Predictions were explicitly anchored to a clinically actionable timepoint (t0, immediately prior to the first scheduled ROP fundus screening), allowing risk stratification and prioritization of ophthalmic screening based solely on information available before the first examination. Through a stepwise feature-selection approach, a concise set of 24 predictors was retained, and the random forest model demonstrated the highest discrimination on the independent internal testing set, supporting its potential utility for early, data-driven risk assessment in this population.

Given the retrospective, single-center design, the observed calibration deviations, and the absence of external validation, the model should be interpreted as a tool for risk ranking within comparable care settings rather than as a universally generalizable prediction rule. Future multicenter prospective studies incorporating leakage-free data preprocessing, model recalibration, and external validation across diverse populations are needed before broad clinical implementation.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12886-026-04798-6.

Supplementary Material 1

Supplementary Material 2

Acknowledgements

The authors thank Huanchen Cao (Information Technology Department) for providing technical support during data extraction. We also thank Hong Li for their assistance with data collection.

Author contributions

XL conceived and designed the study and oversaw the overall project. YY developed the research protocol, collected and analyzed the data, and drafted the manuscript. YZ collected the data, performed the statistical analysis, and contributed to manuscript writing. QD reviewed the study protocol, coordinated the study progress, and critically reviewed the manuscript. YM and HY collected the data. NS and XM assisted with data collection and resource allocation. YX and YG assisted in developing the research protocol and contributed to data analysis. All authors read and approved the final manuscript.

Funding

This study was supported by the Qinghai Province Kunlun Talents Program (High-end Innovation and Entrepreneurship Talent - Leading Talent Project; grant number QHKLYC-GDCXCY-2020-202). The funder was involved in the study design, data collection, analysis, decision to publish, and manuscript review.

Data availability

The datasets used and/or analysed during the current study are not publicly available due to ethical and privacy restrictions, but are available from the corresponding author on reasonable request and with permission of the institutional ethics committee (and, where applicable, a data use agreement).

Declarations

Ethical approval

This study was officially approved by the Ethics Committee of Qinghai Red Cross Hospital (Approval No.: KY-2025-13). All procedures were conducted in accordance with the Declaration of Helsinki and relevant national regulations.

Consent to participate

Given the retrospective nature of this study, the requirement for informed consent from the parents or legal guardians of the participating infants was waived by the Ethics Committee of Qinghai Red Cross Hospital. This waiver was granted in accordance with Chinese national regulations governing biomedical research involving humans (《涉及人的生物医学研究伦理审查办法》, 2016; 《涉及人的生命科学和医学研究伦理审查办法》, 2023), based on the following justifications: (1) the study involved minimal risk as it was a retrospective analysis of de-identified medical records; (2) obtaining informed cosennt was impracticable due to the large sample size and loss of contact with participants after hospital discharge; and (3) strict measures were implemented to protect patient privacy and data confidentiality, including complete de-identification of all personal information.

Consent for publication

Not applicable. The manuscript does not contain any individual person's data in any form (including individual details, images or videos).

Competing interests

The authors declare no competing interests.

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Received: 6 January 2026 / Accepted: 1 April 2026
Published online: 07 April 2026

References

1. Cakir B, et al. Thrombocytopenia is associated with severe retinopathy of prematurity. JCI Insight. 2018;3(19):e124238.
2. Bohley M, et al. A single intravenous injection of cyclosporin A-loaded lipid nanocapsules prevents retinopathy of prematurity. Sci Adv. 2022;8(38):eabo6638.
3. Moin M, et al. Severe ROP rate and assessment of the burden of ROP screening at a single tertiary care public hospital in Pakistan. BMC Ophthalmol. 2025;25(1):594.
4. García H, et al. Global prevalence and severity of retinopathy of prematurity over the last four decades (1985–2021): a systematic review and meta-analysis. Arch Med Res. 2024;55(2):102967.
5. Gundlach BS, et al. Real-world visual outcomes of laser and anti-VEGF treatments for retinopathy of prematurity. Am J Ophthalmol. 2022;238:86–96.
6. Trzcionkowska K, Schalij-Delfos NE, van den Akker-van Marle EME. Cost reduction in screening for retinopathy of prematurity in the Netherlands by comparing different screening strategies. Acta Ophthalmol. 2023;101(1):81–90.
7. Dogra MR, Vinekar A. Role of anti-vascular endothelial growth factor (anti-VEGF) in the treatment of retinopathy of prematurity: a narrative review in the context of middle-income countries. Pediatr Health Med Ther. 2023;14:59–69.
8. Tsai AS, et al. Assessment and management of retinopathy of prematurity in the era of anti-vascular endothelial growth factor (VEGF). Prog Retin Eye Res. 2022;88:101018.
9. Bishnoi K, et al. A narrative review on managing retinopathy of prematurity: insights into pathogenesis, screening, and treatment strategies. Cureus. 2024;16(3):e56168.
10. Shah PK, et al. Retinopathy of prematurity: past, present and future. World J Clin Pediatr. 2016;5(1):35–46.
11. Chen BH. Minimum standards for evaluating machine-learned models of high-dimensional data. Front Aging. 2022;3:901841.
12. Su R, et al. Genomic selection in pig breeding: comparative analysis of machine learning algorithms. Genet Sel Evol. 2025;57(1):13.
13. Ali M, Aittokallio T. Machine learning and feature selection for drug response prediction in precision oncology applications. Biophys Rev. 2019;11(1):31–9.
14. Vannuccini S, et al. Infertility and reproductive disorders: impact of hormonal and inflammatory mechanisms on pregnancy outcome. Hum Reprod Update. 2016;22(1):104–15.
15. Green EA, et al. The role of the interleukin-1 family in complications of prematurity. Int J Mol Sci. 2023;24(3):2815.
16. Fundus Diseases Group in Ophthalmology Branch of Chinese Medical Association. Guidelines for screening retinopathy of prematurity in China in 2014. Chin J Ophthalmol. 2014;50(12):933–5.
17. López-Rueda A, et al. Enhancing mortality prediction in patients with spontaneous intracerebral hemorrhage: radiomics and supervised machine learning on non-contrast computed tomography. Eur J Radiol Open. 2024;13:100618.
18. Willi, Sauerbrei, et al. Selection of important variables and determination of functional form for continuous predictors in multivariable model building. Statist. Med. 2007;26:5512–28.
19. Yu Z. Data-driven discovery of core sleep biomarkers for predicting early cardiometabolic risk in a healthy population using machine learning. medRxiv. 2025.
20. AZUR MELISSAJ, et al. Multiple imputation by chained equations: what is it and how does it work? Int J Methods Psychiatr Res. 2011;20(1):40–9.
21. Xie C, et al. Effect of machine learning re-sampling techniques for imbalanced datasets in (18)F-FDG PET-based radiomics model on prognostication performance in cohorts of head and neck cancer patients. Eur J Nucl Med Mol Imaging. 2020;47(12):2826–35.
22. Demircioğlu A. Applying oversampling before cross-validation will lead to high bias in radiomics. Sci Rep. 2024;14(1):11563.
23. Lin CY, et al. Machine learning-based prediction of three-year heart failure and mortality after premature ventricular contraction ablation. Diagnostics (Basel). 2025;15(21):10281.
24. Elshewey AM, et al. DDoS classification of network traffic in software defined networking SDN using a hybrid convolutional and gated recurrent neural network. Sci Rep. 2025;15(1):29122.
25. Zhou W, et al. Predicting central lymph node metastasis in papillary thyroid microcarcinoma: a breakthrough with interpretable machine learning. Front Endocrinol (Lausanne). 2025;16:1537386.
26. Zhan M, et al. Application of artificial intelligence in conjunction with clinical laboratory indicators to aid decision-making for surgical or conservative treatment of pediatric intestinal obstruction. World J Pediatr Surg. 2025;8(5):e001079.
27. Han H, et al. Development of an interpretable machine learning model to predict short-term bleeding risk in patients receiving dual antithrombotic therapy following cardiac surgery. Int J Clin Pharm. 2025.
28. Heidari P, Milan A. Combining K-fold cross validation with bayesian hyperparameter optimization for accuracy enhancement of land cover and land use classification. Sci Rep. 2025;15(1):39758.
29. Huang H, et al. Predicting rheological properties of asphalt modified with mineral powder: bagging, boosting, and stacking vs. single machine learning models. Mater (Basel). 2025;18(12):2985.
30. Ruiz Sarrias O, et al. Predicting severe haematological toxicity in gastrointestinal cancer patients undergoing 5-FU-based chemotherapy: a Bayesian network approach. Cancers (Basel). 2023;15(17):4278.
31. Kang BY, et al. Serum calcium-based interpretable machine learning model for predicting anastomotic leakage after rectal cancer resection: a multicenter study. World J Gastroenterol. 2025;31(19):105283.
32. Yucel OE, et al. Incidence and risk factors for retinopathy of prematurity in premature, extremely low birth weight and extremely low gestational age infants. BMC Ophthalmol. 2022;22(1):367.
33. Zhang H, et al. Risk factors for retinopathy of prematurity among preterm infants with bronchopulmonary dysplasia. J Matern Fetal Neonatal Med. 2025;38(1):2497058.
34. Wickramasinghe LC, et al. Lung and eye disease develop concurrently in supplemental oxygen-exposed neonatal mice. Am J Pathol. 2020;190(9):1801–12.
35. Rashidian P, Karami S, Salehi SA. A review on retinopathy of prematurity. Med Hypothesis Discov Innov Ophthalmol. 2024;13(4):201–12.
36. Woods J, Biswas S. Retinopathy of prematurity: from oxygen management to molecular manipulation. Mol Cell Pediatr. 2023;10(1):12.
37. Hutchinson AK, et al. Clinical models and algorithms for the prediction of retinopathy of prematurity: a report by the American Academy of Ophthalmology. Ophthalmology. 2016;123(4):804–16.
38. Scott M, Lundberg, et al. Explainable machine-learning predictions for the prevention of hypoxaemia during surgery. Nat BIoMEDIcAL Eng. 2018;10(2):749–60.
39. Raptis S, Ilioudis C, Theodorou K. From pixels to prognosis: unveiling radiomics models with SHAP and LIME for enhanced interpretability. Biomed Phys Eng Express. 2024;10(3):035022.

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